THE UNIVERSITY OF HONG KONG
DEPARTMENT OF MATHEMATICS

MATH2241 Introduction to Mathematical Analysis
Sample Test 2 Solutions

1. (a) For any $\delta > 0$, we have $\frac{\delta}{2} \in (-\delta, \delta) \cap X$ and $\frac{\delta}{2} \neq 0$. So 0 is a limit point by definition.

   (b) We claim that $\lim_{x\to 0} f(x) = -1$. For any $\varepsilon > 0$, we choose $\delta = \sqrt{\varepsilon} > 0$. Then for any $x \in X$ with $0 < |x - 0| < \delta$, we have $|f(x) - (-1)| = \left| x^2 \sin\left(x^3 + \frac{1}{x}\right) \right| \leqslant x^2 < \delta^2 = \varepsilon$. This implies $\lim_{x\to 0} f(x) = -1$.

2. (a) Suppose on the contrary that $\sum_{n=1}^{\infty} \frac{a_n}{a_{n+1}}$ converges. Then $\lim_{n\to\infty} \frac{a_n}{a_{n+1}} = 0$. Hence, there exists $N \in \mathbb{N}$ such that $\left| \frac{a_n}{a_{n+1}} - 0 \right| < 1$ for any $n \geqslant N$. This implies $a_n < a_{n+1}$ for any $n \geqslant N$, and hence the sequence $\{a_n\}$ is increasing for $n \geqslant N$. Thus, $a_n \geqslant a_N > 0$ for all $n \geqslant N$, which contradicts the fact that $\{a_n\}$ has limit 0, as $\sum_{n=1}^{\infty} a_n$ converges.

   (b) Let $a_n = \frac{1}{2^{n^2}}$. Then $\sum_{n=1}^{\infty} \frac{a_{n+1}}{a_n} = \sum_{n=1}^{\infty} \frac{1}{2^{2n+1}}$ converges since it is a geometric series with common ratio $\frac{1}{4} < 1$. Also, the series $\sum_{n=1}^{\infty} a_n$ converges by the comparison test as $a_n \leqslant \frac{1}{2^n}$.

3. It is easy to prove by induction that $b_n > 1$ for any $n \in \mathbb{N}$. Then we have

$$b_{n+1} = \frac{2b_n^2 + b_n + 1}{b_n - 1} = 2b_n + \frac{3b_n + 1}{b_n - 1} > 2b_n.$$

   As $b_1 = 2$, we can prove by induction that $b_n \geqslant 2^n$ for any $n \in \mathbb{N}$. Therefore, $\frac{1}{b_n} \leqslant \frac{1}{2^n}$ for any $n \in \mathbb{N}$. As the geometric series $\sum_{n=1}^{\infty} \frac{1}{2^n}$ converges, $\sum_{n=1}^{\infty} \frac{1}{b_n}$ converges by the comparison test.

   We now compute the partial sums. Note that $b_{n+1} + 1 = \frac{2b_n^2 + b_n + 1}{b_n - 1} + 1 = \frac{2b_n^2 + 2b_n}{b_n - 1}$. So

$$\frac{2}{b_{n+1} + 1} = \frac{b_n - 1}{b_n(b_n + 1)} = -\frac{1}{b_n} + \frac{2}{b_n + 1}.$$

   It follows that $\frac{1}{b_n} = 2\left(\frac{1}{b_n + 1} - \frac{1}{b_{n+1} + 1}\right)$. Therefore, the partial sum is

$$\sum_{n=1}^{k} \frac{1}{b_n} = 2\left(\frac{1}{b_1 + 1} - \frac{1}{b_2 + 1} + \frac{1}{b_2 + 1} - \frac{1}{b_3 + 1} + \cdots + \frac{1}{b_{k-1} + 1} - \frac{1}{b_k + 1}\right) = 2\left(\frac{1}{3} - \frac{1}{b_k + 1}\right).$$

   As $b_k \geqslant 2^k$, the second fraction converges to 0. So the series converges to $\frac{2}{3}$.

4. For any $0 < x < 1$, since $[0, x]$ is a closed and bounded interval and f is continuous, f attains its maximum value on this interval by the extreme value theorem. Hence we can rewrite $g(x) = \max\{f(y) : y \in [0, x]\}$. Note that we must have $g(x) \geqslant f(x)$ by definition, and g is monotonic increasing.

   Suppose $c \in (0, 1)$ and $\varepsilon > 0$ are fixed. Since f is continuous, there exists $\delta > 0$ such that if $x \in [0, 1]$ and $|x - c| < \delta$, then $|f(x) - f(c)| < \dfrac{\varepsilon}{2}$. We claim that $|g(x) - g(c)| < \varepsilon$ for any such x.

   - Firstly, consider those $x \in [0, 1]$ with $c - \delta < x \leqslant c$. If $g(x) = g(c)$, the inequality holds trivially. If $g(x) < g(c)$, suppose $g(c) = f(a)$ for some $a \in [0, c]$. Note that we must have $a \in (x, c]$ by the definition of $g(x)$. Hence $a \in (c - \delta, c]$, and we have

   $$\begin{aligned} |g(x) - g(c)| = g(c) - g(x) = f(a) - g(x) &\leqslant f(a) - f(x) \\ &\leqslant |f(a) - f(c)| + |f(c) - f(x)| < \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon. \end{aligned}$$

   - Similarly, consider those $x \in [0, 1]$ with $c < x < c + \delta$. If $g(x) = g(c)$, the inequality holds trivially. If $g(x) > g(c)$, suppose $g(x) = f(b)$ for some $b \in [0, x]$. Note that we must have $b \in (c, x]$ by the definition of $g(c)$. Hence $b \in (c, c + \delta)$, and we have

   $$|g(x) - g(c)| = g(x) - g(c) = f(b) - g(c) \leqslant f(b) - f(c) < \frac{\varepsilon}{2} < \varepsilon.$$

   Therefore, we have proven if $x \in [0, 1]$ and $|x - c| < \delta$, then $|g(x) - g(c)| < \varepsilon$. This means g is continuous at c.

   Remark. Using the same arguments, one can indeed prove that g is also continuous at 0 and 1, so g is a continuous function.

5. (a) False. Consider $a_n = 0$ for any $n \in \mathbb{N}$. Then the zero series converges, but the series $\displaystyle\sum_{n=1}^{\infty} (-1)^{n+1}$ diverges.

   (b) True. Suppose there exists $\delta > 0$ such that $X \cap (a - \delta, a + \delta)$ is a finite set. Then the set $(X \cap (a - \delta, a + \delta)) - \{a\}$ is also finite. Let this set be $\{b_1, b_2, \ldots, b_k\}$. Note that $k \geqslant 1$ as a is a limit point. Define $d = \min\limits_{1 \leqslant j \leqslant k} \{|a - b_j|\} > 0$. Then $X \cap (a - d, a + d)$ does not contain any b_j. It follows that $X \cap (a - d, a + d)$ is empty or $\{a\}$. In any case, a is not a limit point of X by definition. This is a contradiction.

   (c) False. Define $f(x) = \begin{cases} 0 & \text{if } x = 0, 1, \\ 1 & \text{if } x \neq 0, 1. \end{cases}$ Then f is discontinuous at 0 but $f \circ f$ is the zero function, which is uniformly continuous.